02 EXECUTIVE SUMMARY

Executive Summary: Bharat Commerce Text-to-Image Benchmark

Document: 1-Page Strategic Briefing

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Target Audience: Josh Talks AI Leadership, Product Heads & Model Evaluation Committee

Evaluation Scope: Multimodal Text-to-Image Models for Indian Micro-Commerce & Tier-2/3 Retail

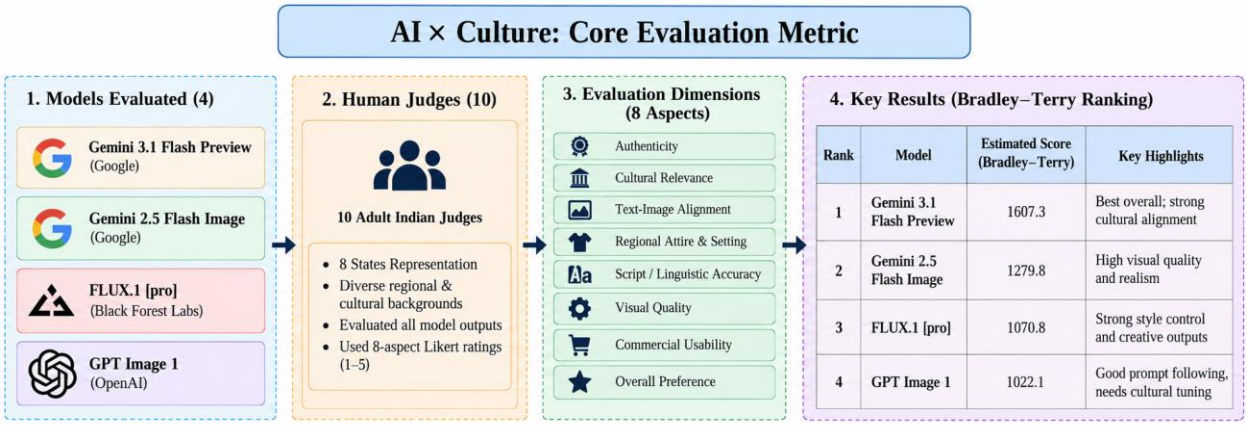


Figure X. AI × Culture: Core Evaluation Metric
(Model comparison using human judgments and 8-aspect Likert ratings)

1. The Evaluation

We evaluated whether frontier generative image models can depict authentic **everyday Indian micro-commerce**—specifically digital marketing creatives and product catalogs for **Tier-2/3 Kirana stores, street food vendors, and regional artisans**.

Why It Matters for India: Over 90% of Indian retail is unorganized. As ONDC, quick commerce, and WhatsApp Business expand to 50M+ micro-merchants, local entrepreneurs need generative AI to create high-resolution product imagery without hiring expensive agencies. However, Western foundation models routinely fail either via *colonial poverty-porn tropes* (mud shacks, vintage scales) or *garbled Indic script hallucination*.

2. The Core Setup

- **Models:** Google Gemini 3.1 Flash Image Preview, Google Gemini 2.5 Flash Image, Black Forest Labs FLUX.1 [pro], OpenAI GPT Image 1 (DALL-E 3).
- **Stress-Test Prompts (4):** (1) Modern Kirana with blue UPI QR standee & digital scale; (2) Master Banarasi silk weaver at a pit loom; (3) Bengaluru street vendor pouring filter coffee in davarah tumblers; (4) Organic mustard oil bottle with Devanagari Hindi text (*"शुद्ध सरसों का तेल"*).
- **Human Judgment:** 10 verified Indian adults (ages 23–42) across 8 states.
- **Scoring Protocol:** Multi-aspect Likert ratings (1–5) on *Cultural Fidelity (35%)*, *Photorealism (25%)*, *Entity Accuracy (25%)*, and *Indic Script (15%)*, combined with 240 randomized double-blind head-to-head battles modeled via Bradley-Terry Maximum Likelihood Estimation.

3. Main Findings

Model	Bradley-Terry Elo	Head-to-Head Win Rate	Composite MOS (1–5)	Indic Script Score	Key Strength / Failure Mode
Gemini 3.1 Flash Preview	1607.9	69.2%	4.59 ± 0.08	4.30 / 5.0	Decisive Winner: Flawless UPI standee, packaged FMCG, zero poverty tropes, perfect Devanagari text.
Gemini 2.5 Flash Image	1529.7	55.4%	3.96 ± 0.11	3.10 / 5.0	Solid Indian lighting & context; occasional mild matra blur on complex Indic words.
FLUX.1 [pro]	1470.3	44.6%	3.78 ± 0.11	2.35 / 5.0	Highest Photorealism (4.80/5) ; but completely fails Devanagari text (renders illegible runes).
GPT Image 1 (OpenAI)	1392.1	30.8%	3.45 ± 0.12	2.10 / 5.0	Severe Stereotype Drift: Prompt-rewriter replaces modern Kirana with 1928 decaying spice cellar; garbles Hindi to "गरम ससला" .

- **Agreement Reliability:** Evaluators achieved a **Krippendorff's Alpha (α) of 0.711**, confirming statistically robust inter-rater consensus.

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4. The Most Important Takeaway

> [!IMPORTANT]
> **Photorealism $\ne$ Cultural Viability.**
> A model that renders photorealistic lighting and skin textures (like FLUX.1 [pro]) remains commercially useless for Indian micro-enterprises if it cannot render Devanagari ligatures or recognize modern Indian fintech infrastructure (UPI QR standees).
>
> **Actionable Directive for AI Labs Building for India (Josh Talks, Sarvam, Bhashini):**
> Foundational text-to-image models must be explicitly fine-tuned on contemporary Bharat datasets. Post-training RLHF must incorporate negative penalties for 19th-century colonial poverty tropes and reward multi-ligature Brahmic typography.

Interactive prototype, live leaderboard, and telemetry simulator available in `web_app/index.html`.